

Terminal Orientation and Survivor Incidence in Five-Token Return Dynamics

One-Lane Dihedral Classification and Pair-Level Non-Descent

WuJun Chen

Independent Researcher

2026

This paper (Paper XXXIV of the RIME program) studies intrinsic raw first-exit incidence in normalized single-defect circular dynamics. It does not assert typed transfer membership, projectability, recursive return, credit settlement, or a reset bound.

Abstract

Problem. Spectator-safe reachability determines which pair of five source lineages can occupy the terminal kernel pair. It does not record where the other three source lineages land after the strict exit. A survivor statement therefore requires more than a marked-pair Safe-Hit witness.

Approach. The construction retains the complete labelled injection and terminal exponent. For a normalized terminal witness (λ, u) , define

$$\beta_{\lambda, u} = \varepsilon_{\Delta} p^u a \lambda.$$

This one map records both the fused source pair and the source-addressed survivor placement. It transports the same witness back to the original raw defect event. The classification then specializes to the one-lane dihedral family $a \in \text{Aut}(C_{n-1})$.

Results. Every dihedral branch has the same five hittable source pairs: the consecutive pairs in the source cyclic order. The survivor spectrum, however, depends on the orientation character. A rotation branch realizes exactly the increasing survivor placements, while a reflection branch realizes both the increasing and decreasing placements. For every hittable pair the spectrum sizes are

$$\binom{n-2}{3} \quad \text{and} \quad 2 \binom{n-2}{3},$$

respectively. Thus equal Safe-Hit and equal fused-pair data do not determine survivor incidence: pair-level reachability forgets the attained terminal orientation class.

Boundary. The orientation spectrum is exact for the one-lane dihedral family. No claim is made that it is a minimal or sufficient survivor quotient for arbitrary branches or multiple lanes. The results remain raw and do not cross the typed transfer or projectability boundary.

Keywords. synchronizing automata; single-defect automata; packet lineages; first-exit incidence; survivor placement; dihedral dynamics; same-witness descent

Notation Table

Symbol	Meaning
$Q = \mathbb{Z}/n\mathbb{Z}$	labelled circular carrier
$p(q) = q + 1$	fixed full cycle
$K = \{0, \Delta\}$	normalized kernel pair
$E = Q \setminus \{0\}$	normalized branch carrier
$a \in \text{Sym}(E)$	branch permutation
ε_Δ	marked collapse $0 \mapsto \Delta$
T_0	five source lineage labels
$\lambda : T_0 \hookrightarrow E$	labelled normalized state
(λ, u)	terminal witness, including terminal rotation u
$\beta_{\lambda, u}$	complete normalized strict-exit incidence
F	fused two-lineage source pair
$R_0 = T_0 \setminus F$	three surviving source lineages
$\mathfrak{F}_a$	fused-pair spectrum
$\mathfrak{S}_a(F)$	source-addressed survivor spectrum over F
ψ	positive cycle on the one-lane carrier E
$\chi(a)$	dihedral orientation character
$\text{OriSpec}_a(F)$	attained terminal-orientation spectrum

1 Introduction

The first-exit problem for five packets has two distinct outputs. The first is the unordered pair of source packets that fuses. The second is the placement of the three packets that survive the strict exit. The first output is a marked-pair reachability question. The second is a source-addressed incidence question.

Paper XXIX retains all five packet lineages and classifies the complete raw frontier for the canonical defect. Papers XXX and XXXI normalize a general binary-kernel rank- $(n-1)$ defect into a marked collapse together with a free branch permutation. Paper XXXII classifies single-lane Safe-Hit for every branch permutation, while Paper XXXIII identifies the same-witness incidence needed once deterministic orbit descent fails [4–8].

Those results answer increasingly broad versions of

$$\text{which pair can fuse?} \tag{1.1}$$

They do not by themselves answer

$$\text{where do the other three source lineages land?} \tag{1.2}$$

A terminal endpoint cannot be reconstructed by choosing one witness for (1.1) and another for (1.2). The pair and survivor placement must come from the same labelled path and the same terminal event.

This paper introduces the normalized incidence map

$$\beta_{\lambda, u} = \varepsilon_\Delta p^u a \lambda. \tag{1.3}$$

Its unique two-element fiber is the fused pair; its restriction to the other three labels is the survivor placement. The terminal exponent u remains part of the witness.

The main classification concerns the one-lane regime $\Delta = 1$. Put $L = n - 1$ and let a be a cycle-graph automorphism of the punctured lane. All such branches have the same complete Safe-Hit predicate, but their survivor spectra split according to the orientation character

$$\chi(a) \in \{+1, -1\}. \tag{1.4}$$

Rotation branches retain only positive terminal orientation. Reflection branches attain both orientations. The fused-pair projection forgets exactly this distinction in the classified family.

The proof has one important converse. A formally allowed terminal placement is not treated as a witness. Because five occupied points in an $n \geq 6$ carrier leave a hole, a terminal rotation can move a hole to the deleted coordinate and reconstruct an actual reachable section injection. This binds fused pair, survivor placement, and terminal exponent to one row.

The theorem spine is:

1. Section 2 defines normalized strict-exit incidence and proves same-witness transport to the original raw event.
2. Section 3 classifies the reachable labelled orientation fibers of every one-lane dihedral branch.
3. Section 4 proves the complete dihedral survivor-spectrum theorem.
4. Section 5 isolates terminal orientation as the information forgotten by pair-level reachability.
5. Section 6 records the computational controls and the boundary to wider branch and typed theories.

1.1 Related Work and Novelty Boundary

The Černý conjecture and the general theory of synchronizing automata concern short reset words [3, 15, 16]. Circular, almost-group, and completely reachable automata provide neighboring structural settings [1, 2, 9, 10, 12, 13]. This paper proves neither a reset bound nor a complete-reachability characterization.

Subset synchronization, monotonicity, and dynamic word constraints study related partial and order-sensitive reachability questions [11, 14, 17]. Here one guarded path must keep five source-labelled lineages injective until one declared strict event, and the output retains the incidence of every source label.

Within the RIME synchronizing-automata line, Paper XXIX proves the canonical source-addressed frontier, Papers XXX–XXXII expose and classify normalized branch dynamics, and Paper XXXIII formalizes source-addressed orbit incidence [4–8]. The new claims here are an exact normalized strict-exit incidence transport, a complete one-lane dihedral survivor-spectrum classification, and a symbolic proof that the pair-level Safe-Hit surface does not determine survivor incidence.

2 Normalized Strict-Exit Incidence

2.1 The inherited branch system

Fix

$$Q = \mathbb{Z}/n\mathbb{Z}, \quad K = \{0, \Delta\}, \quad E = Q \setminus \{0\}, \quad (2.1)$$

with $n \geq 6$. Let

$$\varepsilon_\Delta(0) = \Delta, \quad \varepsilon_\Delta(q) = q \quad (q \neq 0), \quad (2.2)$$

and let $a \in \text{Sym}(E)$. Paper XXXI's branch normalization gives the partial point maps

$$\phi_r = \varepsilon_\Delta p^r a : E \longrightarrow E, \quad r \in \mathbb{Z}/n\mathbb{Z}. \quad (2.3)$$

Let T_0 be a five-element set of source lineage labels. A normalized state is an injection

$$\lambda : T_0 \hookrightarrow E. \quad (2.4)$$

The transition $\lambda \xrightarrow{r} \phi_r \lambda$ is defined exactly when ϕ_r is injective on $\lambda(T_0)$. For a fixed source λ_0 , let

$$\mathcal{L}_{n,\Delta,a}(\lambda_0) \quad (2.5)$$

be the complete set of injections reachable by guarded transitions.

The normalized system is intrinsic. Transport back to an original labelled defect additionally assumes the branch-normalization data of Paper XXXI: a rank- $(n-1)$ defect d with missing image m , $D = Q \setminus \{m\}$, a bijection

$$b = d|_E : E \xrightarrow{\sim} D, \quad p^{t_0}(m) = 0, \quad a = p^{t_0}b.$$

These data are not reconstructed from a . Conclusions internal to $\beta_{\lambda,u}$ need only the normalized system; conclusions about the original raw event are conditional on this supplied realization.

2.2 Terminal witnesses

Fix $F \subset T_0$, $|F| = 2$, and put $R_0 = T_0 \setminus F$. A terminal witness for F is a pair (λ, u) satisfying

$$\lambda \in \mathcal{L}_{n,\Delta,a}(\lambda_0), \quad p^u a \lambda(F) = \{0, \Delta\}. \quad (2.6)$$

Define

$$\boxed{\beta_{\lambda,u} = \varepsilon_{\Delta} p^u a \lambda : T_0 \longrightarrow E.} \quad (2.7)$$

Since $p^u a \lambda$ is injective, (2.7) is equivalently

$$\beta_{\lambda,u}(q) = \begin{cases} \Delta, & q \in F, \\ p^u a(\lambda(q)), & q \in R_0. \end{cases} \quad (2.8)$$

Proposition 2.1 (normalized strict-exit incidence transport).

For every terminal witness (λ, u) , conclusions 1–2 below hold intrinsically. When the normalized system is supplied with the original branch-realization data above, conclusions 3–4 also hold:

1. $\beta_{\lambda,u}$ has the unique two-element fiber F over Δ ;
2. $\beta_{\lambda,u}|_{R_0}$ is an injection into $E \setminus \{\Delta\}$;
3. the one normalized witness transports, without replacement, to the raw witness $(b\lambda, t_0 + u)$ for the original strict defect event; and
4. replacing lineage labels by their fixed source packets recovers the raw rank-four endpoint exactly.

Proof. The only nonsingleton fiber of ε_{Δ} is $\{0, \Delta\}$. Equation (2.6) says that precisely the two lineages in F occupy that fiber. The other three images are distinct and avoid both kernel coordinates, proving the first two claims.

Let $b : E \xrightarrow{\sim} D$ and t_0 be the branch-normalization data of Paper XXXI. The identities

$$b^{-1}d = \varepsilon_{\Delta}, \quad a = p^{t_0}b \quad (2.9)$$

give

$$\boxed{dp^{t_0+u}b\lambda = b\beta_{\lambda,u}.} \quad (2.10)$$

Thus (2.10) carries this one normalized witness to its prescribed raw coordinates $(b\lambda, t_0+u)$; no second witness is chosen. Finally, the fixed source packets are nonempty and disjoint. The output packet at a coordinate is therefore the union of exactly the source packets whose lineage labels occupy that fiber, as in Paper XXIX's lineage decoder.

2.3 Incidence spectra and the same-witness firewall

Let $\mathcal{W}_a(\lambda_0; F)$ be the set of terminal witnesses satisfying (2.6). Define

$$\mathfrak{B}_a(\lambda_0; F) = \{\beta_{\lambda,u} : (\lambda, u) \in \mathcal{W}_a(\lambda_0; F)\}, \quad (2.11)$$

and

$$\boxed{\mathfrak{S}_a(\lambda_0; F) = \{\beta_{\lambda,u}|_{R_0} : (\lambda, u) \in \mathcal{W}_a(\lambda_0; F)\}.} \quad (2.12)$$

The fused-pair spectrum is

$$\mathfrak{F}_a(\lambda_0) = \{F : \mathcal{W}_a(\lambda_0; F) \neq \emptyset\}. \quad (2.13)$$

The attained boundary has the exact decomposition

$$\mathfrak{B}_a(\lambda_0) \cong \bigsqcup_{F \in \mathfrak{F}_a(\lambda_0)} \{F\} \times \mathfrak{S}_a(\lambda_0; F). \quad (2.14)$$

This is a decomposition of attained incidences, not a claim that every formal survivor injection occurs. In particular,

$$\begin{aligned} & [\exists(\lambda, u) \text{ hitting } F] \\ & \quad \wedge [\exists(\lambda', u') \text{ with the desired survivor pattern}] \\ & \not\Rightarrow \exists(\hat{\lambda}, \hat{u}) \text{ realizing both properties.} \end{aligned} \quad (2.15)$$

3 The One-Lane Dihedral System

Set $\Delta = 1$ and $L = n - 1$. On

$$E = \{1, \dots, n - 1\}, \quad (3.1)$$

let ψ be the positive L -cycle. Let

$$\text{Aut}(C_L) = \langle \psi, \rho \rangle, \quad (3.2)$$

where ρ fixes 1 and reverses the cycle. Define

$$\chi(a) = \begin{cases} +1, & a \in \langle \psi \rangle, \\ -1, & a \in \langle \psi \rangle \rho. \end{cases} \quad (3.3)$$

Give T_0 a positive cyclic order and assume that $\lambda_0 : T_0 \hookrightarrow E$ preserves it. An injection is called positive or negative according to whether it preserves or reverses this oriented cyclic order.

Lemma 3.1 (identity-path simulation).

For every $a \in \text{Sym}(E)$, every guarded identity-branch path has a guarded branch- a realization with the same source and target injections.

Proof. The label-zero branch return is $a_{\#}$ and is globally enabled. Since a is a finite permutation, positive powers of this return realize $a_{\#}^{-1}$. Before an identity transition

$$x \mapsto (\varepsilon_1 p^r)_{\#} x,$$

first reach $a_{\#}^{-1}x$ by label-zero returns and then apply the label- r branch return. Its output is

$$(\varepsilon_1 p^r a)_{\#} (a_{\#}^{-1} x) = (\varepsilon_1 p^r)_{\#} x.$$

Iterating this construction simulates the whole path.

Lemma 3.2 (reachable orientation fiber).

If $a \in \text{Aut}(C_L)$, then

$$\mathcal{L}_a(\lambda_0) = \begin{cases} \{\text{positive injections}\}, & \chi(a) = +1, \\ \{\text{positive injections}\} \sqcup \{\text{negative injections}\}, & \chi(a) = -1. \end{cases} \quad (3.4)$$

Proof. By Paper XXXI's lane-order preservation theorem, on every guarded support $\varepsilon_1 p^r$ preserves the hole-deleted cyclic order [4]. A branch step therefore preserves orientation for a rotation branch and reverses it for a reflection branch. This proves the forward inclusions.

Paper XXIX's labelled order-fiber theorem says that the identity branch reaches every positive injection [5]. Lemma 3.1 imports those paths into every branch- a system. If a is a reflection and μ is negative, $a^{-1}\mu$ is positive. Reach $a^{-1}\mu$ and apply the globally enabled label-zero return once to reach μ .

The next lemma is the step that prevents formal terminal coordinates from being promoted silently to actual witnesses.

Lemma 3.3 (missing-hole terminal realization).

Let $a \in \text{Aut}(C_L)$. Suppose that an injective placement $\mu : T_0 \hookrightarrow Q$ is positive when $\chi(a) = +1$, and is either positive or negative when $\chi(a) = -1$. Assume also that

$$\mu(F) = \{0, 1\}. \quad (3.5)$$

Then there is an actual terminal witness (λ, u) with $p^u a \lambda = \mu$.

Proof. The placement occupies five points in an $n \geq 6$ carrier, so it omits some coordinate h . Choose u so that $p^{-u}\mu$ omits 0, and set

$$\boxed{\lambda = a^{-1} p^{-u} \mu : T_0 \hookrightarrow E.} \quad (3.6)$$

If a is a rotation, the allowed μ is positive and hence so is λ . If a is a reflection, Lemma 3.2 contains both orientation classes. Thus λ belongs to the reachable fiber in either case, and $p^u a \lambda = \mu$.

4 Complete Dihedral Survivor Classification

For a consecutive source pair F , orient the source order as

$$(f_0, f_1, r_1, r_2, r_3), \quad F = \{f_0, f_1\}. \quad (4.1)$$

Define

$$\mathfrak{S}^+(F) = \{s : R_0 \hookrightarrow E \setminus \{1\} : 2 \leq s(r_1) < s(r_2) < s(r_3) \leq n-1\}, \quad (4.2)$$

and

$$\mathfrak{S}^-(F) = \{s : R_0 \hookrightarrow E \setminus \{1\} : 2 \leq s(r_3) < s(r_2) < s(r_1) \leq n-1\}. \quad (4.3)$$

Theorem 4.1 (one-lane dihedral survivor-spectrum classification).

For every $a \in \text{Aut}(C_L)$,

$$\boxed{\mathfrak{F}_a(\lambda_0) = \{\text{the five consecutive pairs in the source cyclic order}\}.} \quad (4.4)$$

For every such pair F ,

$$\boxed{\mathfrak{S}_a(\lambda_0; F) = \begin{cases} \mathfrak{S}^+(F), & \chi(a) = +1, \\ \mathfrak{S}^+(F) \sqcup \mathfrak{S}^-(F), & \chi(a) = -1. \end{cases}} \quad (4.5)$$

Consequently,

$$\boxed{|\mathfrak{S}_a(\lambda_0; F)| = \begin{cases} \binom{n-2}{3}, & \chi(a) = +1, \\ 2\binom{n-2}{3}, & \chi(a) = -1. \end{cases}} \quad (4.6)$$

Proof. By Lemma 3.2, the possible pre-collapse terminal placements are positive only for a rotation branch and have both orientations for a reflection branch. An unordered source pair can occupy the adjacent kernel coordinates $\{0, 1\}$ in either orientation exactly when its members are consecutive in the source cyclic order. Lemma 3.3 supplies an actual terminal witness for every such formal placement, proving (4.4).

For a positive terminal placement, (4.1) forces

$$f_0 \mapsto 0, \quad f_1 \mapsto 1, \quad 2 \leq r_1 < r_2 < r_3 \leq n-1. \quad (4.7)$$

For a negative terminal placement it forces

$$f_1 \mapsto 0, \quad f_0 \mapsto 1, \quad 2 \leq r_3 < r_2 < r_1 \leq n-1. \quad (4.8)$$

The strict collapse sends both fused lineages to 1 and fixes every survivor coordinate. Equations (4.7)–(4.8) therefore give (4.5). The two survivor families are disjoint because their coordinates are distinct, and each is indexed by a three-subset of the $n-2$ coordinates $\{2, \dots, n-1\}$. This proves (4.6).

The identity branch is the positive-orientation specialization and recovers Paper XXIX's canonical survivor spectrum after undoing branch coordinates. The reflection branches add one complete negative-orientation copy; they do not change which source pairs can fuse.

5 Terminal Orientation and Pair-Level Non-Descent

For a hittable pair F , define

$$\text{OriSpec}_a(F) = \{\epsilon \in \{+, -\} : \exists(\lambda, u) \in \mathcal{W}_a(\lambda_0; F), \quad p^u a \lambda \text{ has orientation } \epsilon\}. \quad (5.1)$$

Corollary 5.1 (orientation-spectrum factorization).

Within the one-lane dihedral family,

$$\text{OriSpec}_a(F) = \begin{cases} \{+\}, & \chi(a) = +1, \\ \{+, -\}, & \chi(a) = -1, \end{cases} \quad (5.2)$$

and

$$\boxed{\mathfrak{S}_a(\lambda_0; F) = \bigsqcup_{\epsilon \in \text{OriSpec}_a(F)} \mathfrak{S}^\epsilon(F).} \quad (5.3)$$

Proof. This is Theorem 4.1 with the terminal orientation retained instead of counted.

The factorization is exact for the classified family. It does not assert that OriSpec is minimal or sufficient for arbitrary branches.

Corollary 5.2 (pair-level non-descent).

Let a_+ be any rotation and a_- any reflection in $\text{Aut}(C_L)$. Then

$$\mathfrak{F}_{a_+}(\lambda_0) = \mathfrak{F}_{a_-}(\lambda_0), \quad (5.4)$$

and the two branches have the same complete marked-pair Safe-Hit predicate, but for every hittable pair F ,

$$\boxed{\mathfrak{S}_{a_+}(\lambda_0; F) \neq \mathfrak{S}_{a_-}(\lambda_0; F).} \quad (5.5)$$

Hence fused-pair reachability does not determine survivor incidence.

Proof. Equation (5.4) follows from (4.4). Paper XXXII proves that every cycle-graph automorphism has Safe-Hit exactly on the adjacent marked-pair class [6]. Equations (4.5)–(4.6) show that a_+ realizes only $\mathfrak{S}^+(F)$, while a_- also realizes the nonempty disjoint family $\mathfrak{S}^-(F)$.

Corollary 5.2 is a quotient non-descent statement. The pair-level observable forgets terminal orientation. It is stronger than exhibiting two distinct raw endpoints for one fused pair, because the two branch families agree on the entire marked-pair reachability predicate.

6 Controls, Scope, and Open Boundary

6.1 Bounded finite controls

The paper-owned finite control under `experiments/paper34/` retains complete lineage labels and terminal exponents. Its six-point run exhausts all $5! = 120$ branch permutations. It also checks every rotation and reflection of the one-lane cycle for $6 \leq n \leq 9$.

For each dihedral branch and each consecutive fused pair, the control compares the complete attained survivor set, not only its cardinality, against (4.5). The per-pair counts are

n	rotation branch	reflection branch
6	4	8
7	10	20
8	20	40
9	35	70

These bounded computations are consistency controls. They are not the proof of Theorem 4.1 or Corollary 5.2.

6.2 Wider raw branch theory

For a non-dihedral one-lane branch, Paper XXXII shows that every marked-pair orbit is Safe-Hit [6]. That theorem still does not classify the survivor spectrum. An arbitrary branch can alter terminal cyclic order more substantially than the two classes in (5.2). In multiple lanes, lane type, phase incidence, and same-witness mobility may all affect survivor placement.

The next raw problem is therefore not another proof of pair reachability. It is to determine which terminal-order or phase-incidence data classify $\mathfrak{S}_a(F)$ beyond the dihedral family.

6.3 Boundary to typed transfer and recursion

Proposition 2.1 and Theorem 4.1 classify raw incidence. They do not establish that any endpoint belongs to a separately declared transfer fiber. In particular,

$$\text{raw survivor incidence} \not\Rightarrow \text{typed transfer membership} \not\Rightarrow \text{same-witness projectability}. \quad (6.1)$$

No role, ancestry, residual, authorization, or handoff field is inferred from the raw map $\beta_{\lambda,u}$. No new semantic field is introduced to force such a bridge.

7 Computational Artifacts

The paper-owned evidence package is available under [experiments/paper34/](#) in the RIME repository. It has no dependency on the broader exploratory source tree.

Surface	Location	Purpose
Finite control	<code>survivor_incidence_audit.py, results/</code>	bounded same-witness replay
Theorem note	classification note	supporting all- n proof provenance
Lean spine	<code>lean/</code>	incidence and non-descent implications
Validation	<code>validation/</code>	source and evidence gates
Source closure	<code>development-manifest.json</code>	exact 18-artifact paper-owned closure

The finite result is not promoted as proof of the all- n claims. The formal note is supporting proof provenance; the owning theorem statements and claim boundaries are those in this manuscript.

A paper-owned Lean formalization machine-checks the data-independent part of the argument. From one supplied injective terminal placement it derives the unique two-lineage collision fiber, survivor injectivity, and collision-root avoidance. It also checks the missing-coordinate cardinality lemma, the orientation-spectrum count, and the logical non-descent from equal pair data to unequal survivor spectra. The formalization consumes rather than proves the guarded all- n reachability and dihedral orientation classification in Sections 3–4. It does not formalize the finite database or the conditional transport back through a concrete branch conjugacy.

The source closure is verified locally against exact file bytes. Any release receipt remains downstream of those bytes and records its declared validation mode and replay status. It is not an independent mathematical validation.

8 Claim Status and Boundary

Claim surface	Status	Scope
Normalized strict-exit incidence	Proposition 2.1(1)–(2)	every normalized branch and every complete terminal witness
Transport to the original raw event	Proposition 2.1(3)–(4)	every supplied Paper XXXI branch realization of that normalized witness
Reachable orientation fiber	Lemma 3.2	one-lane dihedral branches, all $n \geq 6$
Missing-hole terminal realization	Lemma 3.3	allowed terminal orientations, all $n \geq 6$
Complete dihedral survivor spectrum	Theorem 4.1	every $a \in \text{Aut}(C_{n-1})$, all $n \geq 6$
Orientation-spectrum factorization	Corollary 5.1	one-lane dihedral family
Pair-level survivor non-descent	Corollary 5.2	any rotation branch matched with any reflection branch
Incidence/counting/non-descent implication spine	Machine-checked partial Lean formalization	abstract complete terminal witnesses and supplied positive/negative survivor families
Arbitrary-branch survivor classification	Open	pair Safe-Hit alone is insufficient
Multiple-lane survivor classification	Open	phase incidence and lane mobility retained
Typed transfer and projectability bridge	Open interface theorem	not decided by raw incidence

The paper claims no shortest terminal word, path-length bound, reset threshold, credit settlement, uniform recursive menu, or all-rank descent theorem.

9 Conclusion

The first strict exit carries more information than the identity of the fused pair. Retaining the complete terminal witness produces one normalized incidence map whose unique double fiber records the pair and whose singleton fibers record the three survivors. This map transports exactly to the original raw event.

In the one-lane dihedral family, the missing information is terminal orientation. All rotations and reflections have the same five hittable source pairs and the same marked-pair Safe-Hit predicate. Rotation branches realize one orientation class of survivor placements; reflection branches realize both. Pair-level reachability therefore forgets a genuine source-addressed boundary observable.

This establishes survivor incidence as a new mathematical layer rather than an endpoint count appended to Safe-Hit. Beyond the dihedral family, the task is to identify the phase and orientation data that remain visible at the terminal event while preserving the same witness. Only after that raw problem is solved does the separate typed bridge to transfer membership and projectability become relevant.

References

- [1] Mikhail V. Berlinkov and Cyril Nicaud. Synchronizing almost-group automata. *International Journal of Foundations of Computer Science*, 31(8):1091–1112, 2020.
- [2] David Fernando Casas Torres. Completely reachable almost group automata, 2024.

- [3] Ján Černý. Poznámka k homogénnym experimentom s konečnými automatmi. *Matematicko-fyzikálny časopis Slovenskej Akadémie Vied*, 14:208–216, 1964.
- [4] WuJun Chen. Branch-normalized partial returns in single-defect circular automata. RIME Paper XXXI, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23050809>.
- [5] WuJun Chen. Cyclic lineage dynamics and rank-five first-exit frontiers. RIME Paper XXIX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23028361>.
- [6] WuJun Chen. Five-token dynamics beyond lane-order symmetry. RIME Paper XXXII, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23053169>.
- [7] WuJun Chen. Source-addressed stabilizer descent in five-token return dynamics. RIME Paper XXXIII, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23056470>.
- [8] WuJun Chen. Universal return groups and punctured rotations. RIME Paper XXX, version 1.0, Zenodo, September 2026. DOI: <https://doi.org/10.5281/zenodo.23034428>.
- [9] Henk Don. The Černý conjecture and 1-contracting automata. *The Electronic Journal of Combinatorics*, 23(3):P3.12, 2016.
- [10] Lucien Dubuc. Sur les automates circulaires et la conjecture de Černý. *RAIRO Informatique théorique et applications*, 32:21–34, 1998.
- [11] David Eppstein. Reset sequences for monotonic automata. *SIAM Journal on Computing*, 19(3):500–510, 1990.
- [12] Robert Ferens and Marek Szykuła. Recognizing completely reachable automata in quadratic time. *ACM Transactions on Algorithms*, 22(2):1–36, 2026.
- [13] François Gonze and Raphaël M. Jungers. On completely reachable automata and subset reachability. In *Developments in Language Theory*, volume 11088 of *Lecture Notes in Computer Science*, pages 330–341. Springer, 2018.
- [14] Andrew Ryzhikov and Anton Shemyakov. Subset synchronization in monotonic automata. *Fundamenta Informaticae*, 162(2–3):205–221, 2018.
- [15] Mikhail V. Volkov. Synchronizing automata and the Černý conjecture. In *Language and Automata Theory and Applications*, volume 5196 of *Lecture Notes in Computer Science*, pages 11–27. Springer, 2008.
- [16] Mikhail V. Volkov. Synchronization of finite automata. *Russian Mathematical Surveys*, 77(5):819–891, 2022.
- [17] Petra Wolf. Synchronization under dynamic constraints. In *40th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2020)*, volume 182 of *Leibniz International Proceedings in Informatics*, pages 58:1–58:14. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020.